



Statistical Modeling of Stellar Evolution

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Core Objective & Data Strategy

Testing a 70-year-old prediction: Does stellar mass control lifetime?

With Gaia DR3 we built a cosmic laboratory:

- 100k stars within 1 kpc — astrometric precision ($\text{RUWE} < 1.4$, $\sigma_{\varpi}/\varpi < 0.2$)
- Evolutionary phases classified via ML clustering (K-means on HR diagram)
- Mass proxies from T_{eff} ($M \propto T^2$) and luminosity ($M \propto L^{1/3.5}$)

Data enables first empirical test of mass-lifetime scaling at galactic scale.

Theoretical Foundation

The main-sequence lifetime is determined by fuel supply and consumption rate:

$$\tau_{\text{MS}} \propto \frac{M}{L}, \quad L \propto M^{3.5} \Rightarrow \tau_{\text{MS}} \propto M^{-2.5}$$

Hertzsprung-Russell Diagram Analysis

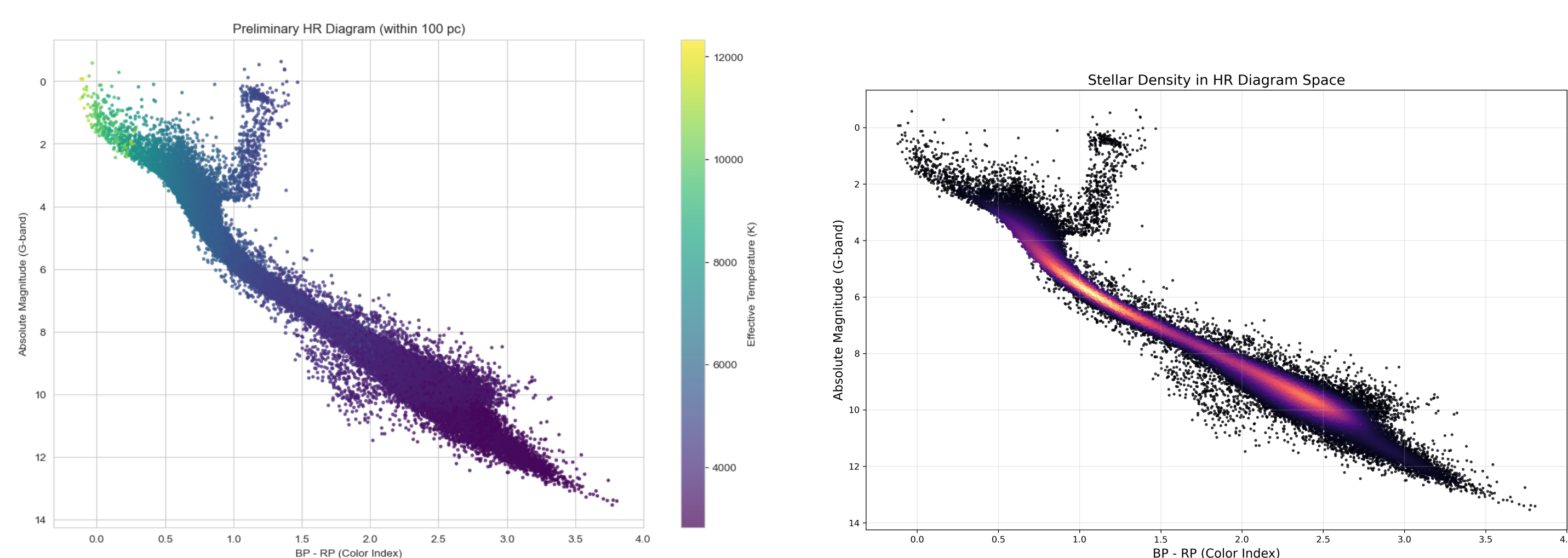


Figure 1. **Left:** Temperature-colored HR diagram revealing main evolutionary sequences. **Right:** Density map highlighting stellar concentrations.

Mass Estimation Methods

From Effective Temperature:

$$\frac{M}{M_{\odot}} = \left(\frac{T_{\text{eff}}}{5772} \right)^2$$

From Stellar Luminosity:

$$\frac{M}{M_{\odot}} = \left(\frac{L}{L_{\odot}} \right)^{1/3.5}, \quad \text{where } \frac{L}{L_{\odot}} = 10^{(4.74 - M_G)/2.5}$$

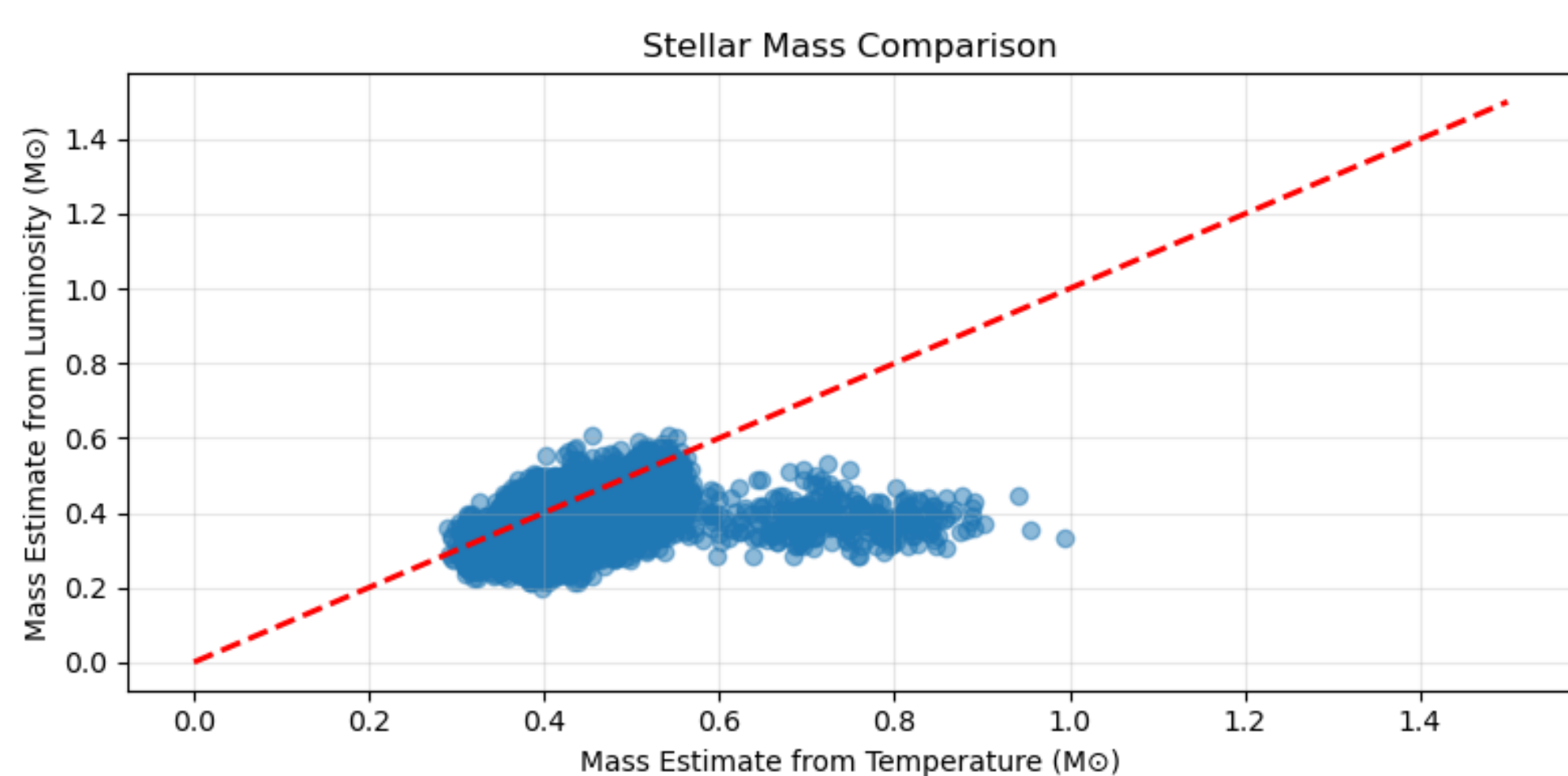


Figure 2. Excellent agreement between mass estimates from temperature vs. luminosity validates our approach.

Evolutionary Classification via K-means Clustering

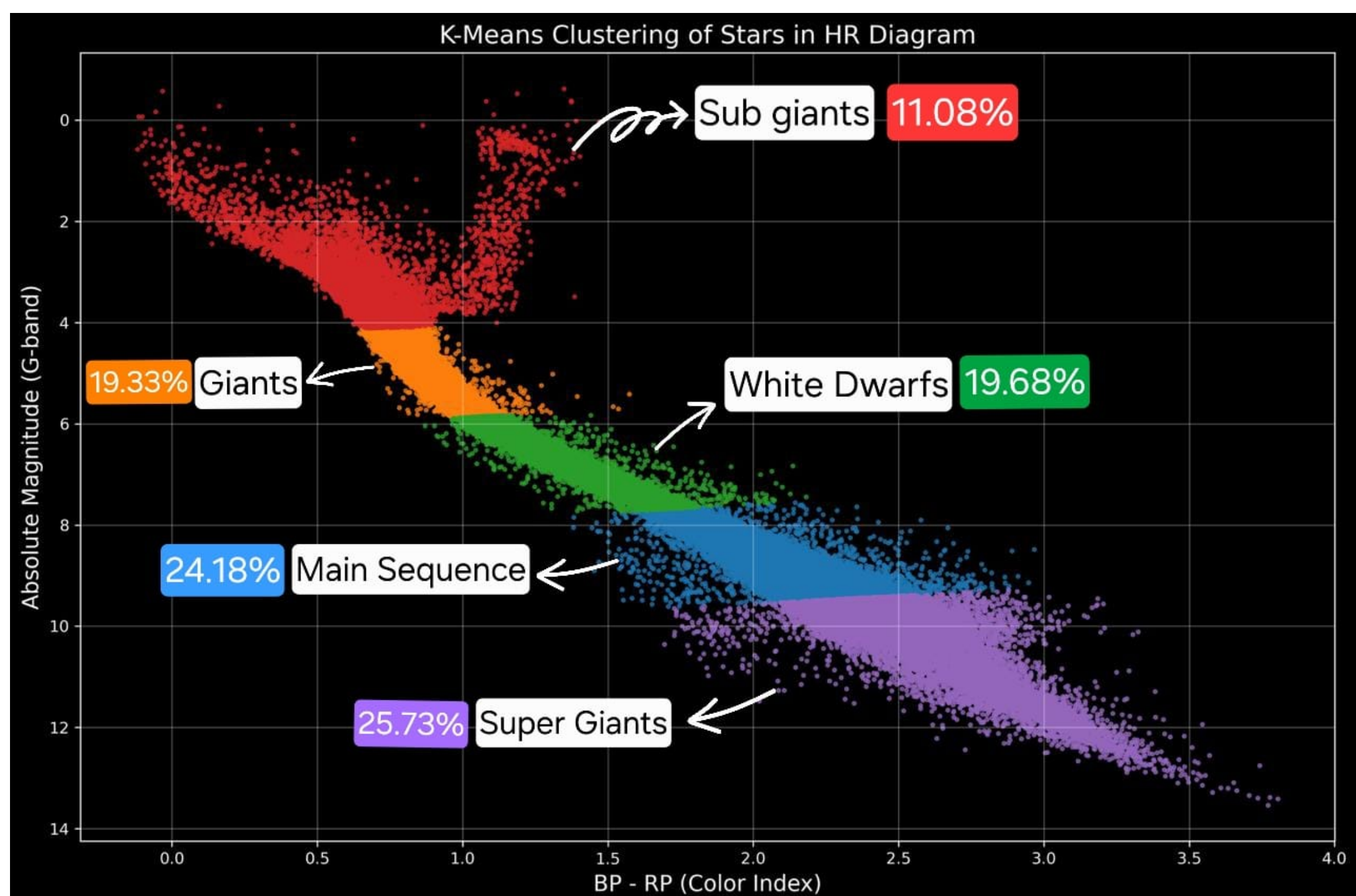


Figure 3. K-means clustering in the HR diagram reveals five distinct evolutionary phases.

Statistical Method 1: Kaplan-Meier Survival Analysis

Non-parametric survival function:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i} \right)$$

where d_i = events (leaving main sequence) at time t_i , n_i = stars at risk

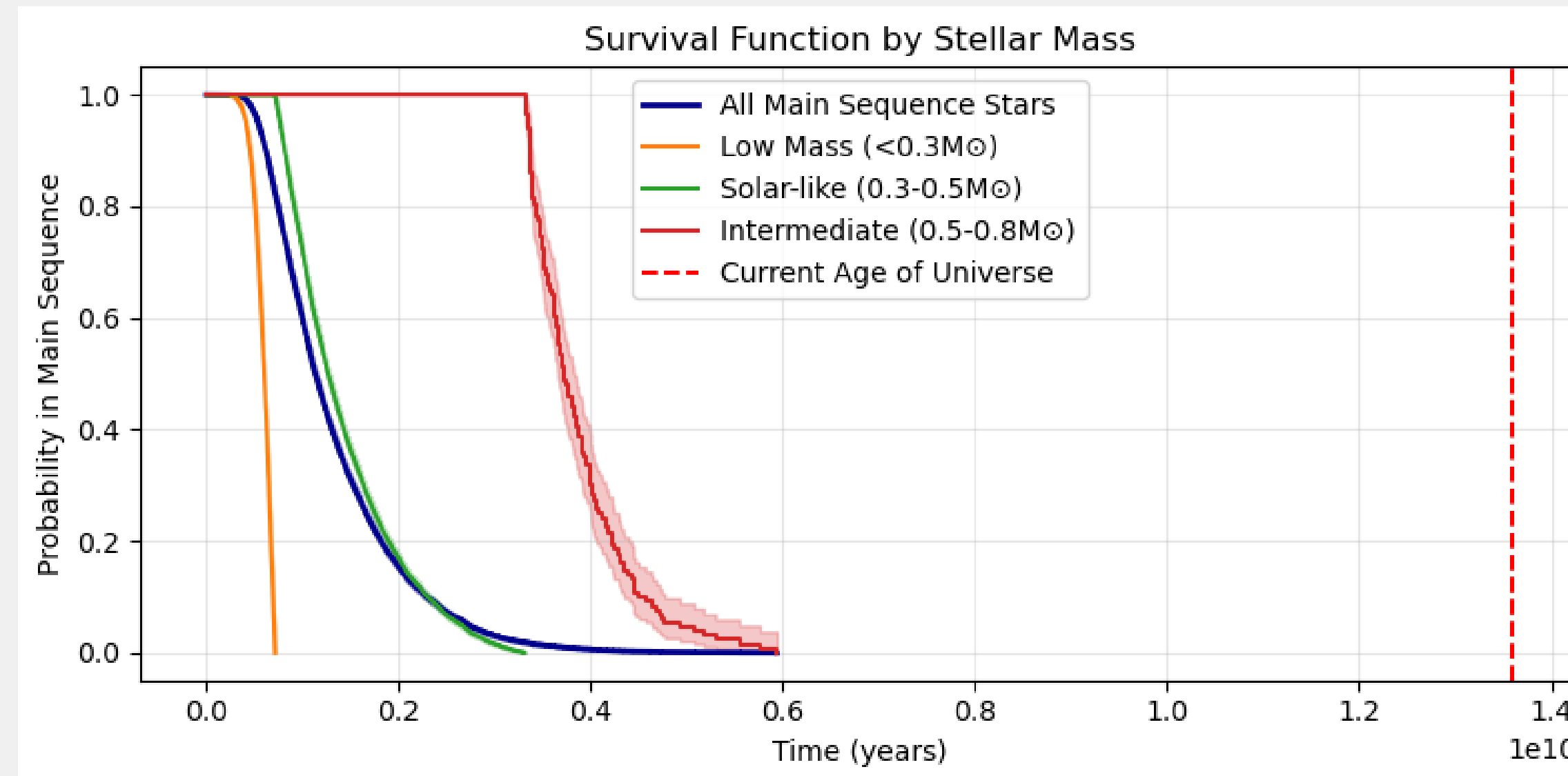


Figure 4. Kaplan-Meier survival curves for four mass bins. Higher-mass stars show faster decline, confirming shorter lifetimes.

Statistical Method 2: Weibull Accelerated Failure Time Model

Parametric survival model:

$$\log T = \beta_0 + \beta_1 \log M + \beta_2 T_{\text{eff}} + \sigma \epsilon$$

Parameter	Value (95% CI)	Interpretation
Mass (β_1)	-2.66 (-2.68, -2.64)	Lifetime $\propto M^{-2.66}$
λ (scale)	1.19×10^{10}	Char. lifetime ~ 10 Gyr
ρ (shape)	1.98 (1.97, 1.99)	Increasing failure rate
T_{eff}	0.00 (0.00, 0.00)	No additional power

Table 1. Weibull AFT model results. Concordance = 0.92, AIC = 429,318.

Statistical Method 3: Bayesian Hierarchical Power Law Model

Full probabilistic inference:

$$\tau(M) = a \cdot M^b$$

Censored likelihood approach:

- On main sequence: $\tau > \text{age}$ (right-censored) & Evolved off MS: $\tau < \text{age}$ (left-censored)

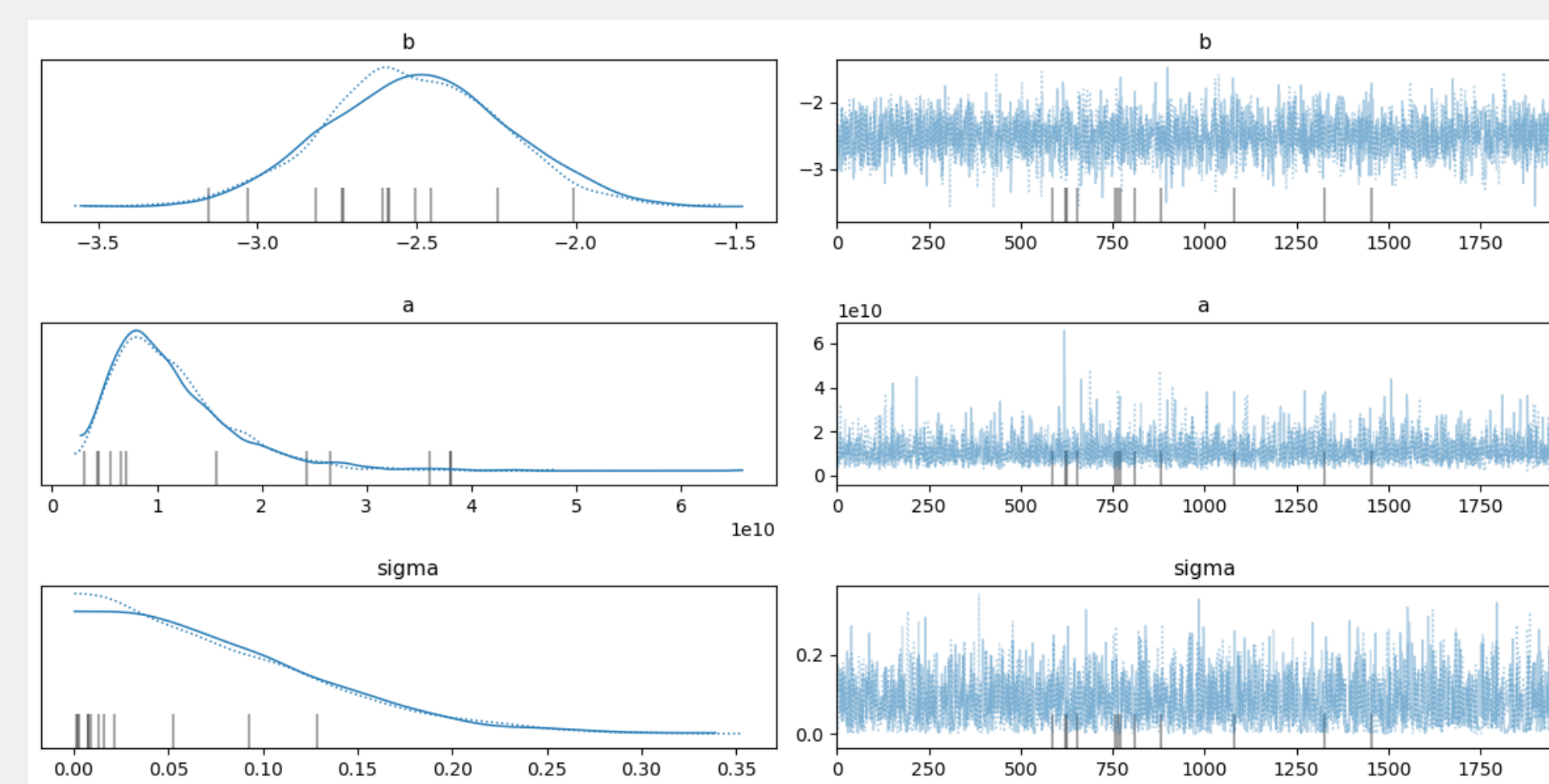


Figure 5. Posterior distributions for power law parameters from MCMC sampling.

Parameter	Mean	SD	95% HDI
b (power law index)	-2.51	0.29	[-3.04, -1.97]
a (normalization)	1.13×10^{10}	6.06×10^9	$[2.82, 22.6] \times 10^9$
σ (scatter)	0.079	0.060	[0.000, 0.184]

Table 2. Bayesian model results with excellent convergence ($\hat{R} = 1.0$, ESS > 1600).

References

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Complete Project code and details

